

B6.1 Monitoring & Maintaining the Environment

OCR A GCSE Biology (J247) Gateway — Topic B6: Global challenges

1. Biodiversity

Biodiversity = the variety of living organisms in an ecosystem. It includes the variety of species, the genetic variety within each species, and the variety of ecosystems present.

High biodiversity makes ecosystems more stable and resilient.

Biodiversity matters because:

- Ecosystems with more species are more stable — better at resisting disease or environmental change
- Provides essential ecosystem services (pollination, water purification, soil formation, oxygen, climate regulation)
- Source of medicines (many drugs come from plants, fungi, microbes)
- Cultural/aesthetic value

2. Sampling Techniques

Tool	Use
Quadrat	Sampling plant or sessile (non-moving) animal populations — count organisms within a square of known area
Transect	Investigating change along an environmental gradient (e.g. distance from a pond)

Pooter	Collecting small invertebrates (e.g. ground beetles) by sucking them into a jar
Sweep net	Collecting flying/grass-dwelling insects by sweeping through vegetation
Pitfall trap	Trapping ground-dwelling invertebrates — small container buried with rim level with the ground
Mark-release-recapture	Estimating mobile animal populations

Mark-release-recapture (Lincoln Index)

Capture, mark, release a sample of organisms. Wait for them to mix back into the population, then capture a second sample. Estimate population using:

$$N = (n_1 \times n_2) \div m$$

where N = total population, n_1 = first sample (marked), n_2 = second sample, m = recaptured marked individuals.

Worked example

A scientist marks 50 woodlice. After release, they capture 60 woodlice; 10 are marked. Estimate the population.

$$N = (50 \times 60) \div 10 = \mathbf{300 \text{ woodlice.}}$$

Assumptions: no births/deaths/migration; marking does not affect survival; marked individuals mix randomly with population.

3. Indicator Species and Pollution Monitoring

An **indicator species** is a species whose presence (or absence) indicates the level of

pollution. They are sensitive to specific pollutants.

Pollution	Indicator species	Indicates
Air (sulfur dioxide)	Lichens	Many lichen types = clean air; few or none = polluted air
Air (SO ₂)	Blackspot fungus on roses	Present = clean; absent = polluted (because SO ₂ kills the fungus)
Water (low O ₂ / sewage)	Mayfly nymphs / stonefly nymphs	Present = clean, oxygenated water
Water (high pollution)	Bloodworms / sludgeworms	Present = polluted water (low O ₂)

Non-living monitoring: scientists also use sensors to measure pH, temperature, dissolved O₂, nitrate, particulates, CO₂, etc. directly.

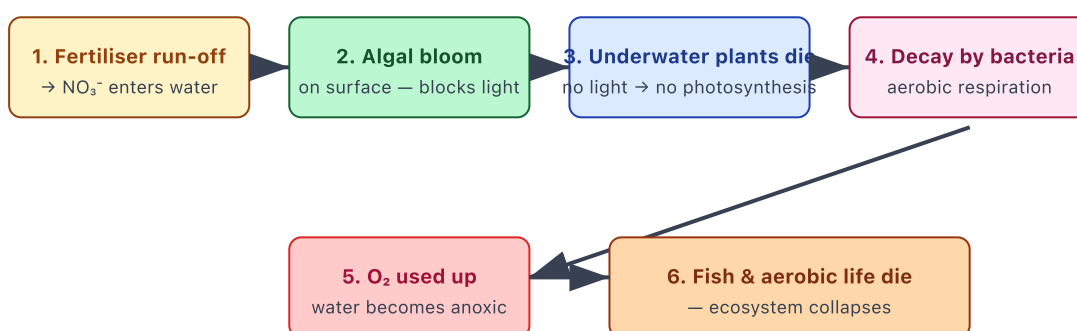
4. Human Impact on Biodiversity

Impact	Cause	Effect
Deforestation	Land cleared for farming, building, fuel	Habitat loss; reduced biodiversity; CO ₂ released; soil erosion
Habitat destruction	Urban expansion, agriculture, mining	Loss of species; food chains disrupted
Pollution (air, water, land)	Industry, transport, agriculture, plastic waste	Death of sensitive species; eutrophication; bioaccumulation
Climate change	CO ₂ emissions, methane (cattle, landfills)	Habitat shifts; coral bleaching; melting ice; species extinctions

Overexploitation	Overfishing, hunting, poaching, illegal trade	Species decline; ecosystem disruption
Invasive species	Introduction of non-native species (often by humans)	Outcompete native species; can cause extinctions

Eutrophication

Eutrophication — Killing a Pond



Climate change

Increased CO₂, methane and other greenhouse gases trap infrared radiation in the atmosphere, raising average temperatures (the **greenhouse effect**). Effects include:

- Melting polar ice → rising sea levels
- More extreme weather (storms, droughts, floods)
- Shifting species ranges (some can't adapt → extinction)
- Coral bleaching from warming ocean and acidification (more dissolved CO₂)

5. Conservation

Conservation = active management of the environment to protect species and ecosystems.

Method	How it helps
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Breeding programmes (in zoos)	Increase numbers of endangered species — release back into wild
Seed banks	Preserve genetic material of plants — useful if a species goes extinct in the wild
Protected areas (national parks, marine reserves)	Habitat preserved; hunting/fishing restricted; species protected by law
Reintroduction programmes	Captive-bred individuals released to restore populations
Reducing deforestation / replanting	Restores habitat; locks up carbon
Education and ecotourism	Raises awareness; provides funding for conservation
Legal protection (CITES)	Bans trade in endangered species

Trade-offs: conservation may conflict with farming, building, business interests. Some argue protecting nature is essential; others prioritise economic development. Solutions often involve compromise (e.g. sustainable forestry, reserves with buffer zones).

6. Sustainable Fisheries

Overfishing threatens fish populations and the wider ecosystem. Sustainable fishing techniques aim to harvest fish in a way that does not deplete the population.

Method	How it works
Net mesh size limits	Larger mesh lets young/small fish escape so they can reproduce
Fishing quotas	Limit total weight of each species caught per year
Closed seasons	No fishing during spawning periods
Marine protected areas	No-fishing zones allow populations to recover

7. Required Practical Link

Linked PAGs include: **using a quadrat to estimate population size** in a habitat (e.g. dandelions on a school field), **using a transect** to investigate distribution along an environmental gradient. See PAG notes for full method.